Shreyash Kumar

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Education

Bachelor of Technology, Electrical Engineering 2021 - 2025
Minors: Computer Systems, Machine Learning and Applications, Philosophy
Indian Institute of Technology Kanpur | GPA: **8.84**/10 *Kanpur, India*

Relevant Coursework: Computer Systems Security, Parallel Computing, Embedded and Cyber-Physical Systems, Digital Communication Networks, Principles of Database Systems, Data Structures and Algorithms, Fundamentals of Computing, Digital Electronics, Deep Learning and Computer Vision, Introduction to Machine Learning, Probability and Statistics

Research Experience

CISPA Helmholtz Center for Information Security Dortmund, Germany
Research Intern *May 2025 – Jul. 2025*

- Investigated novel side-channel leakage vectors in the Linux kernel, targeting context-switch paths and user interrupts
- Reverse-engineered the Intel user-interrupt (UIPI) framework on Sapphire Rapids CPUs using recompiled kernel builds
- Probed UIPI handling for race conditions, sending interrupts into the unregister window to test MSR state safety
- Built a frequency-fingerprinting channel timing a custom ISR to read a neighboring core's frequency from user space

Indian Institute of Technology Kanpur Kanpur, India
Undergraduate Researcher, Cache-Timing Attacks on PQC | Prof. Angshuman Karmakar *Aug. 2024 – Apr. 2025*

- Studied and implemented Flush+Reload and Prime+Probe cache side-channel attacks on the shared L3 cache
- Explored the Mastik toolkit and reproduced a published Flush+Reload attack on the HQC post-quantum scheme
- Reviewed LESS literature and designed a Prime+Probe attack on its secret-indexed Fisher–Yates permutation shuffle
- Established that leaked swap indices fully reconstruct the secret permutation, collapsing an $N!$-size key space

DCode Lab, Indian Institute of Technology Kanpur Kanpur, India
Research Intern *May 2023 – Jul. 2023*

- Developed a bounded, data-driven guidance law for intercepting highly maneuvering, accelerating targets in 2-D
- Formulated guidance as a Markov Decision Process and trained a DDPG policy mapping target states to guidance laws
- Validated the policy across simulations, intercepting 100% targets under randomized square-wave acceleration profiles

Professional Experience

Goldman Sachs Hyderabad, India
Software Engineer *Jul. 2025 – Present*

- Architected AWS workflows orchestrating fund-onboarding across 6+ services, cutting a multi-day process to ∼60s
- Implemented service-to-service authentication using mTLS and AWS STS temporary credentials across internal services
- Built a cached reporting pipeline with a daily precompute job, cutting report latency from 24 hours to 2 minutes
- Designed a recursive Java/SQL aggregation API computing fee-rate discounts over nested fund-transfer hierarchies

C3iHub, IIT Kanpur Kanpur, India
Student Intern (Part-time) *Aug. 2024 – Oct. 2024*

- Accessed a car's infotainment system via the OBD-II port to sniff and analyze CAN frames for vehicle diagnostics
- Designed and developed a Python/Qt-based GUI to visualize, interpret, and interact with CAN frame data in real time
- Reverse-engineered CAN bus frames, enabling remote control of various car functions using the MCP2515 controller

Jaguar Land Rover Bengaluru, India
Software Engineer Intern *May 2024 – Jul. 2024*

- Simulated a multi-ECU Ethernet network in CANoe, scripting spoofing, tampering, and replay attacks using CAPL
- Deployed MACsec (802.1AE) with integrity, replay, and confidentiality protection, blocking 100% of injected attacks
- Developed a secure workflow to provision cryptographic keys and certificates from backend services into vehicle ECUs

Projects

Haar-Accelerated Deepfake Detection System Jan. 2025 – Apr. 2025
Course Project | Prof. Koteswar Jerripothula *IIT Kanpur*

- Replaced CNN-based face detection with an OpenCV Haar cascade pipeline, cutting preprocessing time by nearly 5×
- Enhanced the ResNeXt–LSTM model using squeeze-and-excitation blocks and a residual classifier for feature learning
- Trained and evaluated the model on a 2,000-video FaceForensics++ split, achieving 76.5% deepfake-detection accuracy

An Online Approach to the Dynamic Vehicle Routing Problem Aug. 2024 – Dec. 2024
Course Project, Cyber-Physical Systems | Prof. Indranil Saha *IIT Kanpur*

- Applied reinforcement learning for real-time paratransit routing, achieving 100% request fulfilment with 5 vehicles
- Benchmarked a Markov-chain routing model against a least-squares baseline, showing better efficiency and accuracy
- Tuned Monte Carlo Tree Search depth and runtime cutoff to a 30 s decision deadline without degrading route quality

Parallelizing Stencil Computation Feb. 2024 – Apr. 2024
Course Project | Prof. Preeti Malakar *IIT Kanpur*

- Optimized MPI-based halo exchange for 5-point and 9-point stencil computation, enhancing performance by 30%
- Incorporated topology-aware hierarchical communication, enhancing performance in multi-process conditions by 25%
- Benchmarked the program on the HPC2010 cluster to demonstrate and validate job-scheduling heuristics

Technical Skills

- **Languages**: C, C++, Python, Java, SQL, Bash, TypeScript, JavaScript, Verilog HDL
- **Systems & Tools**: Linux kernel modules, MPI, perf, gdb, QEMU/KVM, Mastik, CANoe/CAPL, AWS, Git
- **Frameworks & Libraries**: PyTorch, OpenCV, NumPy, Spring Boot, React, JUnit, Mockito, Snowflake

Honors and Awards

- **Top 0.5%**, All India Rank 711 among 165K+ candidates in JEE Advanced 2021
- **Top 0.1%**, All India Rank 791 among 1.3M+ candidates in JEE Main 2021
- **KVPY Fellowship** by IISc Bangalore, awarded twice (All India Rank 832 in SA, All India Rank 1227 in SX) 2019, 2021
- **Students' Gymkhana All-Rounder Award**, IIT Kanpur, for academic and extracurricular distinction 2025
- **Top 1% nationally**, Indian Olympiad Qualifier in Chemistry 2021

Leadership & Activities

Institute Secretary, Games and Sports Council Apr. 2024 – Apr. 2025
IIT Kanpur *Kanpur, India*

- Led a three-tier team of 16 coordinators and 100+ secretaries towards executing events reaching 9,000+ students
- Achieved 30% growth in women's representation in the council and 20% growth in graduate participation across events
- Founded the institute's E-Sports Club, organizing 5+ events in its inaugural tenure and securing ∼$2K in sponsorships
- Led gym expansion worth ∼$230K and initiated indoor arena and swimming pool projects worth ∼$165K combined

Men's Volleyball Team 2022 – 2025
IIT Kanpur *Kanpur, India*

- Represented IIT Kanpur at the 55th, 56th, and 57th Inter IIT Sports Meets; won gold at Udghosh '22 among 15+ teams
- Organized workshops, camps and training sessions to increase engagement within the campus community
- Received the Special Appreciation Award (2024–25) for promoting games and sports at IIT Kanpur

Volunteering and Mentoring
IIT Kanpur *Kanpur, India*

- **Prayas**: Mentored and taught children from under-privileged families near the IIT Kanpur campus
- **Counselling Service**: Student Guide to 6 incoming freshmen. Helped organize the UG orientation for 1200 students
- **Vox Populi**: Researched and wrote articles as Assistant Editor in a 50+ member student journalism team
- **Animal Welfare Cell**: Conducted awareness sessions and aided in the vaccination of 10+ stray dogs